Lecture7(B)

JavaScript DOM

Summary of the previous

- What is JavaScript?
- Embedding JavaScript with HTML
- JavaScript conventions
- Variables in JavaScript
- JavaScript operators
- Input output in JavaScript
- JavaScript functions
- Conditional Statements
- Looping Statements

Outlin

- **Dialog boxes in JavaScript**
- **HTML Document Object Model (DOM)**

1. Dialog boxes in JavaScript

- JavaScript provides the ability to pickup user input or display small amounts of text to the user by using dialog boxes
- These dialog boxes appear as separate windows and their content depends on the information provided by the user

1.1 Alert

- **An alert box is simply a small message box that pops up and gives the user some information**
- **An alert dialog box is mostly used to give a warning message to the users**
- **When an alert box pops up, the user will have to click "OK" to proceed**
- **Syntax:**
  - **alert(“message”)**

1.1 Alert B

alert - Notepad

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```
<script language="javascript">
function alertme(message)
{
alert(message)
}
alertme("Welcome to my page")
</script>
</head>
<body>
<h1> Hello World!</h1>
</body>
</html>
```

Start of the function

Displays an alert box

Calling the function

1.2 Prompt

- A **prompt box** is often used if you want the user to **input** a value before entering a page
- When a prompt box pops up, the user will have to click either **"OK"** or **"Cancel"** to proceed after entering an input value
- If the user clicks "OK" the box returns the **input value**
- If the user clicks "Cancel" the box **returns null**

1.2 Prompt

prompt - Notepad

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```
<html>
<head>
<title>Prompt box</title>
<script language="javascript">
function promptMe()
{
var name=prompt("please Enter Your Name",
"Name");
document.write("Hello Mr. " + name)
}
promptMe()
</script>
</head>
```

Start of the function

Prompt box

Calling the function

1.3 Confirm

- A confirm box is often used if you want the user to verify or accept something
- When a confirm box pops up, the user will have to click either "OK" or "Cancel" to proceed
- If the user clicks "OK", the box returns true
- If the user clicks "Cancel", the box returns false

1.3 Confirm

confirm - Notepad

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```
<head>
<title>Confirm Box</title>
<script language="javascript">
function conName()
{
var name=prompt("Your Name please!")
if(confirm("Write on the webpage ?"))
document.write("Welcome Mr." + name)
else
document.write("Welcome MR. Anonymous")
}
conName()
</script>
```

User input

confirmation

Not confirmed

2. Document Object Model (DOM)

- **Once html element are rendered (painted) in the browser window, browser can not recognize them**
- **To create interactive web pages it is vital that the browser continues to recognize HTML elements**
- **JavaScript enabled browsers do this because they recognize and uses DOM**

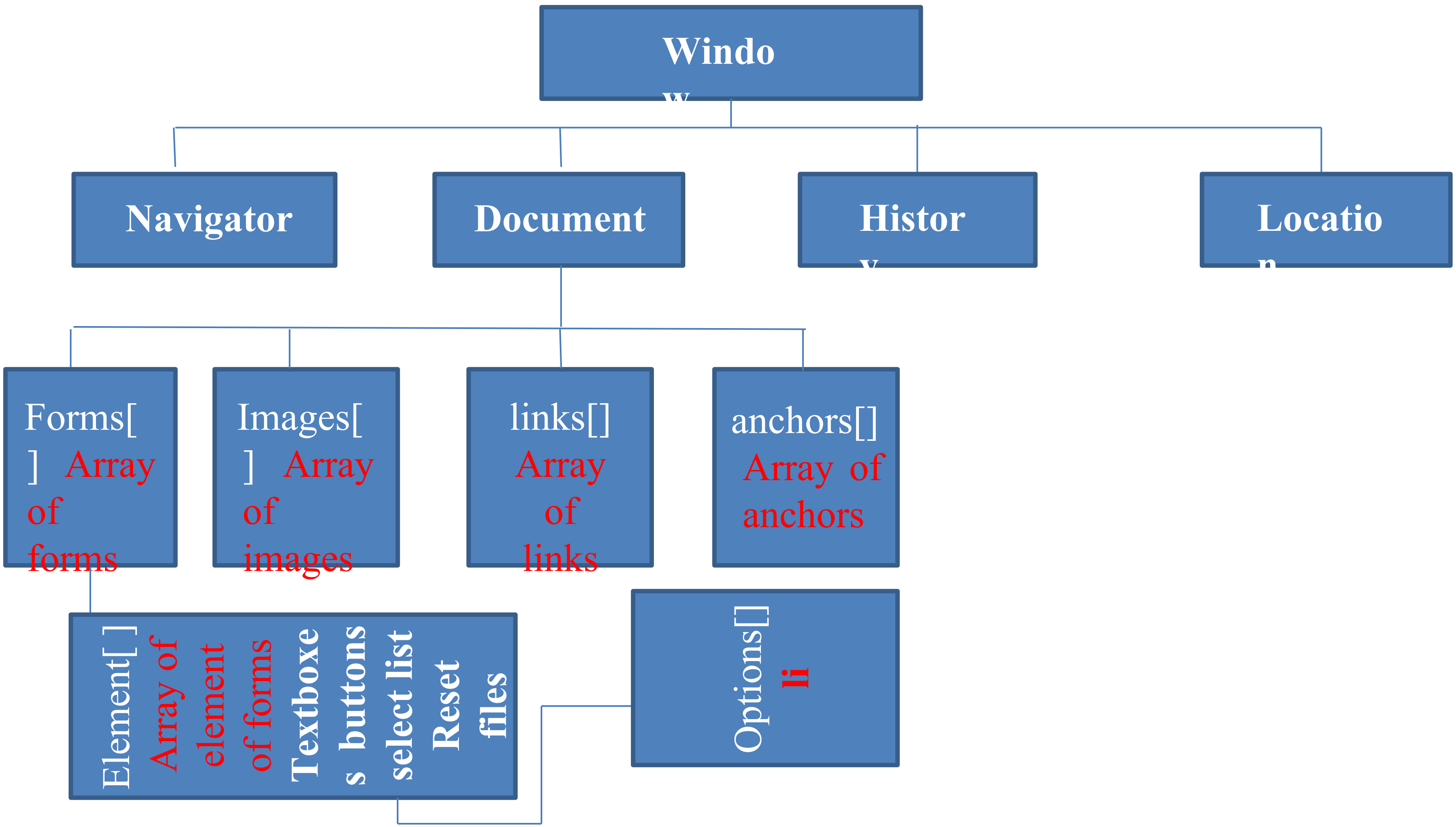
2. Document Object Model (DOM)

- The **HTML DOM** defines a standard set of objects for HTML, and a standard way to **access and manipulate** HTML documents
- **All HTML elements**, along with their containing text and attributes, can be **accessed through the DOM**
  - The contents can be **modified or deleted**, and new elements can be created

2. Document Object Model (DOM)

- The **HTML DOM** is platform and language Independent
  - It can be used by any **programming language** like Java, JavaScript, and VBScript
- The HTML DOM can be thought of as a **hierarchy** moving from the most **general** object to the **most specific**

2. Document Object Model (DOM)



2. Document Object Model (DOM)

- document.forms[0].elements[0].value
- document.images[0].src
- document.links[0].href

2.1 Retrieving HTML

- **The getElementById() method is a workhorse method of the DOM**
- **It retrieves a specified element of the HTML document and returns a reference to it**
- **To retrieve an element, it must have an unique id**
  - **document.getElementById("element-id")**

2.1 Retrieving HTML

- The returned reference can be used to retrieve element attributes
  - document.getElementById("element- id").attribute
  - document.getElementById("pic").src
  - document.getElementById("pic").height

2.1 Retrieving HTML

getElement - Notepad

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```
<title>Retrieving values</title>
<script>
function getValue()
{
var a=document.getElementById("pic")
alert(a.title)
alert(a.height)
}
</script>
</head>
<body>
<img src="bg1.jpg" id="pic" height="100"
title="My favorite car" onClick="getValue()">
```

Function starts

Getting reference to pic

Accessing Attributes

Image tag

Setting id

Calling function

2.2 Retrieving the text of an

- innerHTML property defines both the HTML code and the text that occurs between that element's opening and closing
  - document.getElementById(“element-id”).innerHTML

2.2 Retrieving the text of an

inner - Notepad

File Edit Format View Help

```
<script>
function getText()
{
var a=document.getElementById("mypara")
alert(a.innerHTML)
}
</script>
</head>
<body>
<p id="mypara" onClick="getText()">
Hello World </p>
</body>
</html>
```

Getting reference to element

Getting text

Id of paragraph

Calling the function

Paragraph text

2.3 Getting value of

- **getAttribute()** **method is used to retrieve values of attributes**
  - **document.getElementById("element-id").getAttribute("attribute-name")**
  - **document.getElementById("pic").getAttribute("src")**

2.3 Getting value of

getatt - Notepad

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```
<script>
function getAtt()
{
var a=document.getElementById("pic")
alert(a.getAttribute("title"))
}
</script>
</head>
<body>
<img src="bg1.jpg" id="pic" height="100"
title="My favorite car" onClick="getAtt()">
</body>
</html>
```

Getting reference to element

Using getAttribute function

2.4 Setting value of

- **setAttribute()** **method is used to set values of attributes**
  - **document.getElementById("element-id").setAttribute("attribute-name", "Value")**
  - **document.getElementById("pic").setAttribute("src", "abc.jpg")**

2.4 Setting value of

```
<script>
function setAtt()
{
var a=document.getElementById("pic")
a.setAttribute("src","bg3.jpg")
}
</script>
</head>
<body>
<img src="bg1.jpg" id="pic" height="100"
title="My favorite car" onClick="setAtt()">
</body>
</html>
```

Setting Reference to element

Changing the value of attribute

Summar

- **Dialog boxes in JavaScript**
- **HTML DOM**
- **Retrieving HTML elements**
- **Setting HTML elements**

Reference

- **Chapter 11.** Beginning HTML, XHTML,CSS, and JavaScript, by Jon Duckett, Wiley Publishing; 2009, ISBN: 978-0-470-54070-1.
- **Chapter 3,6,** Learn JavaScript, by Chuck Easttom, Wordware Publishing; 2002, ISBN 1-55622-856-2